

CASE 1 – API Automation & Security Testing

Objective

The objective of this test suite is to validate the functionality, security, and robustness of the POST /api/v1/sandbox/auth endpoint used for Sandbox Authentication.

Tools Used

- Java
- REST Assured
- TestNG
- Maven

Functional Test Coverage

Scenario: Successful Authentication

Validated authentication using a valid request payload containing:

- api_key
- mock_id_number
- device_data

Validations Performed

- Verified HTTP status code is 200 OK
- Verified response contains a valid auth_token
- Verified token is not null or empty
- Verified API response structure

Security Edge Case Testing

Replay Attack Validation

To simulate replay attack behavior, the exact same authentication payload was submitted twice within 100 milliseconds.

Expected Behavior

The API should reject duplicate authentication attempts and return:

- 401 Unauthorized
- OR
- 409 Conflict

Result

The security validation confirmed that duplicate rapid requests are properly rejected.

Token Tampering Validation

Generated a valid authentication token and manually modified the token before reusing it in the Authorization header.

Expected Behavior

The API should reject manipulated or invalid tokens with:

- 401 Unauthorized

Result

The API correctly rejected the tampered token request.

Test Data Management Strategy

- Request payloads were externalized into utility classes/configuration files.
- Dynamic test data generation approach was followed for mock ID numbers.
- This improves maintainability and scalability.